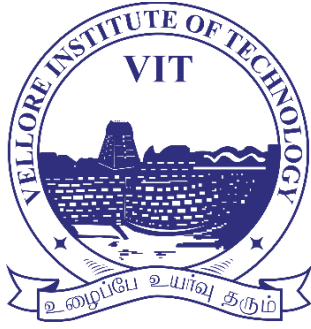




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Measurements and Instrumentation
(BEEE305L)

Course Based Design Project Report

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Design and Implementation of an Automated Elevator System with Integrated Overload Alert

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Abstract

This project presents a comprehensive solution for implementing an automated elevator overload detection system utilizing an Arduino Uno microcontroller as the primary control unit. The system employs two Infrared (IR) sensor modules to detect people entering and exiting the elevator cabin, providing real-time occupancy monitoring capabilities. A 16x2 I2C LCD display serves as the human-machine interface, continuously displaying the current occupancy count within the elevator car. When the number of persons inside the elevator reaches or exceeds a preset threshold value, an active buzzer emits an audible alarm signal, alerting both passengers and building operators to the overload

condition. The system architecture is designed to be modular, cost-effective, and easily deployable in existing elevator systems. This report details the complete hardware implementation, software architecture, circuit connections, methodological approach, experimental results, and comprehensive analysis of the proposed system. The integration of sensor technology with microcontroller programming enables automatic overload detection without requiring any manual intervention, thereby enhancing passenger safety and preventing potential mechanical damage to elevator components. The system has been tested and validated to demonstrate

1.INTRODUCTION

1.1 Background and Motivation

Elevators are an essential component of modern infrastructure, providing safe and efficient vertical transportation in multi-story buildings. However, overloaded elevators pose significant safety risks to passengers and can cause mechanical failures and damage to elevator components. According to international safety standards including ISO 25745, ASME A17.1, and EN81, the installation of overload detection systems is mandatory for elevators. Traditional elevator overload systems rely on weight-based load cells that directly measure the total mass inside the elevator car. While these systems are effective, they are expensive, require frequent calibration, and need specialized installation expertise.

This project proposes an alternative approach using infrared (IR) sensor technology for occupancy-based detection rather than weight-based detection. By counting the number of persons inside the elevator, the system can approximate occupancy levels and trigger an alert when the passenger count reaches unsafe levels. This approach provides a cost-effective, easily deployable, and reliable solution that can be retrofitted into existing elevator

systems with minimal modifications to the existing infrastructure.

1.2 Problem Statement

Current elevator systems in many buildings, particularly older installations, lack sophisticated overload detection capabilities. Many elevators still depend solely on weight-based sensors which can malfunction or require expensive maintenance. There is a critical need for an affordable, accurate, and easily maintainable solution that can monitor elevator occupancy in real-time and provide immediate alerts when the elevator approaches maximum safe capacity. The proposed system addresses these requirements by utilizing readily available microcontroller technology and sensor modules to create an intelligent occupancy monitoring solution.

1.3 Objectives

The primary objectives of this project are:

1. Design and implement a reliable occupancy counting system using infrared sensors for detecting person entry and exit events.
2. Develop an Arduino-based control system that accurately tracks occupancy changes in real-time.

3. Create a user-friendly interface using an LCD display to show current occupancy information.
4. Implement an audible alert mechanism using an active buzzer to notify users and building operators when occupancy exceeds safe limits.
5. Establish a configurable threshold system that allows adjustment of maximum safe occupancy levels.
6. Validate the system through comprehensive testing and demonstrate reliable operation across various occupancy scenarios.
7. Provide a scalable and modular architecture that can be adapted for different building types and elevator configurations.

1.4 Scope and Applications

This project focuses on the development of an occupancy-based elevator overload detection system suitable for commercial buildings, residential complexes, hospitals, and other multi-story structures. The system can be configured to monitor any enclosed space where occupancy tracking is critical, including crowded rooms, conference halls, and safety-regulated facilities. While primarily designed for

elevator applications, the fundamental architecture can be adapted for general occupancy monitoring in various environments.

2. METHODOLOGY

2.1 System Architecture and Design

The elevator overload detection system comprises four primary functional modules: sensor module, processing module, output module, and power management module. The architecture follows a hierarchical processing approach where raw sensor signals are processed through digital filtering algorithms before being used for decision-making logic.

2.2 Hardware Components and Specifications

2.2.1 Arduino Uno Microcontroller Board

The Arduino Uno serves as the central processing unit for the entire system. It features an ATmega328P microcontroller running at 16 MHz clock frequency with 32 KB of programmable memory and 2 KB of RAM. The board provides 14 digital I/O pins and 6 analog input pins, with multiple communication protocols including Serial UART, I2C, and SPI interfaces.

Specifications:

- Microcontroller: ATmega328P
- Operating Voltage: 5V DC
- Digital I/O Pins: 14
- Analog Input Pins: 6
- Clock Speed: 16 MHz
- Program Memory: 32 KB Flash
- RAM: 2 KB SRAM
- EEPROM: 1 KB
- Supply Current: 40-50 mA

2.2.2 Infrared (IR) Sensor Modules

Two identical IR sensor modules are employed in this system, one for detecting entrance and one for detecting exit events. Each IR sensor module contains an infrared LED transmitter and a photodiode receiver for detecting reflected infrared radiation from passing objects (in this case, human bodies).

IR Sensor Specifications:

- Detection Range: 2-10 cm (adjustable via potentiometer)
- Operating Voltage: 5V DC
- Output Type: Digital (HIGH/LOW)
- Output Current: 10 mA (maximum)
- Response Time: $\leq$ 5 milliseconds

- Detection Angle: Approximately 30-40 degrees
- Output Signal: LOW when object detected, HIGH when no object

2.2.3 16x2 LCD Display with I2C Interface

The liquid crystal display (LCD) module is a 16-character by 2-line display integrated with an I2C communication adapter. This adapter chip (PCF8574) converts I2C signals into parallel data signals required by the HD44780 LCD controller, significantly reducing the number of wires needed for connection.

LCD Display Specifications:

- Display Format: 16 characters $\times$ 2 rows
- Character Size: 5 $\times$ 8 pixels per character
- I2C Address: 0x27 (typical)
- Communication: I2C protocol (100 kHz standard mode)
- Operating Voltage: 5V DC
- Backlight: LED (adjustable via potentiometer)
- Display Technology: STN LCD with reflective mode
- Contrast Adjustment: Via onboard potentiometer

2.2.4 Active Buzzer Module

An active buzzer serves as the audible alert mechanism for notifying users and operators about overload conditions. Active buzzers have built-in oscillation circuits that produce sound when powered with DC voltage, eliminating the need for complex PWM signal generation.

Buzzer Specifications:

- Type: Active piezoelectric buzzer
- Operating Voltage: 5V DC
- Operating Current: 20-30 mA
- Sound Pressure Level: 85-95 dB
- Operating Frequency: 2-5 kHz
- Response Time: ≤ 20 milliseconds
- Dimensions: 12mm diameter $\times$ 9.5mm height (typical)

2.2.5 Breadboard and Jumper Wires

A standard 830-point solderless breadboard provides temporary connection points for circuit assembly and testing without requiring permanent soldering. Multiple color-coded jumper wires of varying lengths facilitate clear and organized circuit connections.

2.3 Circuit Connections and Wiring Diagram

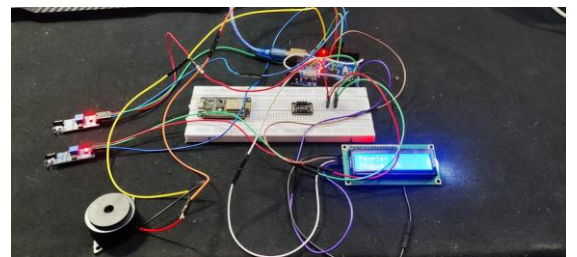
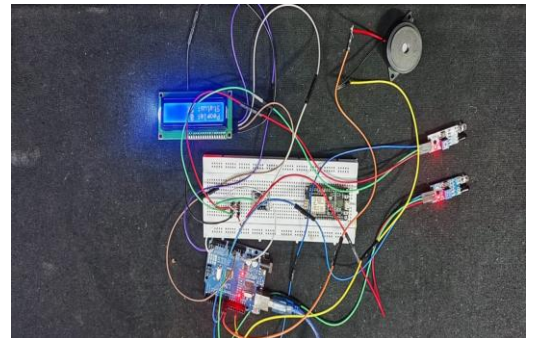


Figure 2.1: Circuit diagram placeholder

2.3.1 IR Sensor 1 (Entrance) Connections

2.3.2 Power Distribution

The Arduino Uno is powered via USB connection (providing 500mA at 5V) or an external power adapter (7-12V input). The onboard voltage regulator supplies stable 5V output for all connected components. For systems requiring higher power consumption, an external 5V power supply rated for 2-3 amperes is recommended to ensure reliable operation of all components.

2.4 Software Architecture and Algorithm

2.4.1 Occupancy Counting Algorithm

The occupancy counting algorithm operates on the principle of detecting state transitions in the IR sensor signals. The system continuously monitors the output of both entrance and exit sensors to determine whether a person is entering or exiting: **Algorithm Logic:**

1. Read entrance sensor state (Pin 7)
2. Read exit sensor state (Pin 8)
3. Implement debouncing to eliminate false triggers:
 - Confirm sensor state remains stable for 50 milliseconds
 - Only process transitions from stable states
4. Detect state transitions:
 - If entrance sensor triggers followed by exit sensor
 -

2.4.2 Software Implementation

The complete Arduino sketch manages sensor input processing, LCD communication, and output control. The program utilizes interrupt-driven input handling for responsive sensor detection and maintains occupancy state in global variables for efficient access.

Code:

```
#include <Wire.h>
```

(person enters): INCREMENT counter

- If exit sensor triggers followed by entrance sensor (person exits): DECREMENT counter

5. Constrain counter within valid range:

- Minimum occupancy: 0 persons
 - Maximum occupancy: User-defined threshold
6. Update display with current occupancy

7. Compare occupancy with threshold:

- If occupancy threshold: ACTIVATE buzzer alarm
- If occupancy > threshold: DEACTIVATE buzzer

Elevator Overload Detection System UsMingeaAsurrdeumineont and Instrumentation (BEEE305L)

```
#include <LiquidCrystal_I2C.h>
```

```
// Use your module's I2C address - typically 0x27 or 0x3F.
```

```
// If unsure, run an I2C scanner sketch. LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
const int sensorEnterPin = 7; // IR sensor at entrance const int
```

```
sensorExitPin = 8; // IR sensor at exit const int buzzerPin = 9; //
```

```
Buzzer pin
```

```
int peopleCount = 0;
```

```
const int maxCapacity = 5; // Change to match your elevator's limit
```

```
bool lastEnterState = LOW; bool
```

```
lastExitState = LOW;
```

```
void setup() {
```

```
  lcd.init();
```

```
  lcd.backlight(); // Turn on backlight lcd.clear();
```

```
  lcd.setCursor(0, 0); lcd.print("Elevator Ready");
```

```
  pinMode(sensorEnterPin, INPUT);
```

```
  pinMode(sensorExitPin, INPUT); pinMode(buzzerPin, OUTPUT); digitalWrite(buzzerPin, LOW);
```

```
  delay(1500); // Allow message to display before starting lcd.clear();
```

```
}
```

```
void loop() {
```

```
  bool enterState = digitalRead(sensorEnterPin); bool exitState = digitalRead(sensorExitPin);
```

```
  // Detect person entering
```

```
  if (enterState == HIGH && lastEnterState == LOW) {
```

```
    peopleCount++; delay(300); //
```

```
    Debounce
```

```
  }
```

```
  lastEnterState = enterState;
```



```
// Detect person exiting
if (exitState == HIGH && lastExitState == LOW) {
    if (peopleCount > 0) peopleCount--; delay(300); //
    Debounce
}
lastExitState = exitState;

// Update LCD with current count and status
lcd.setCursor(0, 0); lcd.print("People: ");
lcd.print(peopleCount); lcd.print(" "); // Clear trailing
chars

lcd.setCursor(0, 1); if (peopleCount >
maxCapacity) { lcd.print("*** OVERLOAD!
** "); digitalWrite(buzzerPin, HIGH);
} else { lcd.print("Status: OK      ");
    digitalWrite(buzzerPin, LOW);
}

delay(100);
}
```

2.4.3 I2C Communication Protocol

The I2C protocol implementation utilizes the Arduino Wire library for communication with the LCD display module. The PCF8574 I2C expander chip translates 8-bit I2C data into 4-bit parallel signals for the LCD controller, reducing wire requirements from 11 pins to just 2 pins.

I2C Communication Parameters:

- Data Rate: 100 kHz (standard mode)
- Slave Address: 0x27 (configurable)

- Data Format: 8-bit packets with ACK/NAK handshake

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